

Cristina M. Aggazzotti

✉ caggazz1@jhu.edu | 🏠 caggazzotti.github.io | 💻 caggazzotti | 🌐 c-aggazzotti

Education

Harvard University

PH.D. IN LINGUISTICS

M.A. IN LINGUISTICS

Cambridge, MA

2019

2017

University College London

M.Sc. IN COGNITIVE AND DECISION SCIENCES (Distinction)

London, UK

2013

University of Southern California

B.A. IN APPLIED AND COMPUTATIONAL MATHEMATICS

B.A. IN LINGUISTICS

Magna Cum Laude; Renaissance Scholar

Los Angeles, CA

2011

Research and Work Experience

Assistant Research Scientist

HUMAN LANGUAGE TECHNOLOGY CENTER OF EXCELLENCE, JOHNS HOPKINS UNIVERSITY

Baltimore, MD

Oct. 2024 – Present

- Develop methods for authorship attribution (who wrote something) and speaker attribution from speech transcripts (who said something), examining how someone's language use can reveal or protect (anonymization) their identity
- Work toward robust and explainable attribution models by exploring model representations, biases, and performance

Postdoctoral Fellow in Computer Science

HUMAN LANGUAGE TECHNOLOGY CENTER OF EXCELLENCE, JOHNS HOPKINS UNIVERSITY

Baltimore, MD

Aug. 2023 – Oct. 2024

- Established a new benchmark and state-of-the-art performance for speaker attribution—identifying a speaker based on the content of their speech (using syntax/semantics/lexicon)—as a supplement to speaker recognition methods, which analyze the speech signal (using acoustics/phonetics)
- Identified the need for speaker anonymization methods to obfuscate both the speech signal and speech content
- Resulted in a first author TACL journal publication and an acceptance for oral presentation at ACL 2024

Postdoctoral Fellow in Linguistics

UNIVERSITÉ DU QUÉBEC À MONTRÉAL

Remote

May 2022 – July 2023

- Developed and tested stylometric and neural authorship attribution models across genres, domains, and dataset sizes
- Identified forensic linguistic data needs and began developing standards for multidisciplinary data use
- Resulted in 3 forensic science and forensic linguistics conference presentations and 2 papers (in preparation)

Academic Editor and Proofreader

FREELANCER

Remote

Feb. 2017 – July 2023

- Edited and proofread journal papers, books, job/grant applications, and dissertations for non-native English speakers

Visiting Research Scholar (Summer Camp for Applied Language Exploration)

HUMAN LANGUAGE TECHNOLOGY CENTER OF EXCELLENCE, JOHNS HOPKINS UNIVERSITY

Baltimore, MD

Jun. 2022 – Aug. 2022

- Experimented with schemas masking training data to explain neural author models' reliance on style versus content
- Developed a linguistic and stylometric feature model for determining authorship of various text types
- Resulted in a first author TACL journal publication and an acceptance for oral presentation at EMNLP 2023

Research Contractor

INSTITUTE FOR LINGUISTIC EVIDENCE

Remote

2016 – 2018; Oct. 2021 – Apr. 2022

- Compared discriminant analysis (SPSS) and LSTM (Python) approaches to classifying suicide notes and last words
- Analyzed linguistic evidence across authors and text genres; researched statistical methods for presentation in court
- Resulted in 4 forensic science conference oral presentations and leading a half-day conference workshop

Research Scientist and Customer Experience Engineer

POINTMAIL, INC. (STARTUP)

Cambridge, MA

Feb. 2018 – Jan. 2020

- Investigated ways to improve phrase- and sentence-level text prediction based on individual language usage and style
- Led bi-weekly team-wide research meetings and helped guide future research directions and product decisions
- Built rapport with customers, onboarded users, resolved customer issues, and elicited & evaluated customer feedback

Graduate Researcher

HARVARD UNIVERSITY

Cambridge, MA

Aug. 2015 – May 2019

- Created a unified analysis of reflexives and reciprocals, implemented in synchronous tree adjoining grammar (a syntax-semantics hierarchical tree pairing system), and provided a novel extension of the formalism to capture morphology
- Developed a syntactic and semantic annotation scheme for reflexives using the Corpus of Contemporary American English by creating a hybrid classification system based on previous reflexives research and attested data
- Mentored 7 undergrads in annotation schemes, machine learning of anaphor data, and testing my model using OCaml

Graduate Researcher

UNIVERSITY COLLEGE LONDON

London, UK

Sep. 2012 – Sep. 2013

- Created a formalized crime script of cash-in-transit robbery combining influence diagrams, causal models, Bayesian networks, and scripts that could ultimately be used by the police to identify effective prevention strategies

Publications and Presentations

- **Multimodal speaker verification as a threat to speaker anonymization.**
Ashi Garg, Cristina Aggazzotti, Leibny Paola García-Perera, and Nicholas Andrews.
Under review (2026).
- **STEB: Style Text Embedding Benchmark.**
Rafael Rivera Soto, Anna Wegmann, and Cristina Aggazzotti.
Under review (2026).
- **A survey on representing linguistic style: Challenges and opportunities.**
Anna Wegmann, Cristina Aggazzotti, Rafael Rivera Soto, and Dong Nguyen.
Under review (2026).
- **A stylometric analysis of speaker attribution from speech transcripts.**
Cristina Aggazzotti and Elizabeth Allyn Smith.
In revision (2026).
- **Ground truth in computational forensic linguistics: Semantic variation and the evaluation of scientific evidence.**
Elizabeth Allyn Smith and Cristina Aggazzotti.
Submitted (2026).
- **CommonLID: Re-evaluating state-of-the-art language identification performance on web data.**
Pedro Ortiz Suarez, Laurie Burchell, Catherine Arnett, ..., Cristina Aggazzotti, ..., Kenton Murray, and Sarah Luger
Proceedings of the Association for Computational Linguistics (2026).
- **Content anonymization for privacy in long-form audio.**
Cristina Aggazzotti, Ashi Garg, Zexin Cai, and Nicholas Andrews.
Proceedings of the IEEE International Conference on Acoustics, Speech, and Signal Processing (2026). Oral presentation.
- **The impact of automatic speech transcription on speaker attribution.**
Cristina Aggazzotti, Matthew Wiesner, Elizabeth Allyn Smith, and Nicholas Andrews.
Transactions of the Association for Computational Linguistics 13, pp. 1578-1596 (2025). Presented at EMNLP 2025.
- **The relationship between linguistic accommodation and speaker attribution.**
Cristina Aggazzotti and Elizabeth Allyn Smith.
5th Focus Conference on Forensic Linguistics of the International Language and Law Association (ILLA) (2025).
- **Speaker attribution of speech transcripts: A stylometric approach.**
Cristina Aggazzotti, Elizabeth Allyn Smith, and Nicholas Andrews.
17th Biennial Conference of the International Association for Forensic and Legal Linguistics (IAFLL) (2025).
- **Weaknesses of artificial intelligence proxies of topic for forensic linguistic applications.**
Cristina Aggazzotti and Elizabeth Allyn Smith.
Proceedings of the American Academy of Forensic Sciences Annual Conference, vol. 31, p. 505 (2025).
- **Can authorship attribution models distinguish speakers in speech transcripts?**
Cristina Aggazzotti, Nicholas Andrews, and Elizabeth Allyn Smith.
Transactions of the Association for Computational Linguistics 12, pp. 875–891 (2024). Presented at ACL 2024.
- **Can authorship representation learning capture stylistic features?**
Andrew Wang*, Cristina Aggazzotti*, Rebecca Kotula, Rafael Rivera Soto, Marcus Bishop, and Nicholas Andrews.
Transactions of the Association for Computational Linguistics 11, pp. 1416–1431 (2023). Presented at EMNLP 2023.

- **Testing authorship analysis on spoken language transcripts: Establishing a first benchmark.**
Cristina Aggazzotti, Nicholas Andrews, and Elizabeth Allyn Smith.
4th Focus Conference on Forensic Linguistics of the International Language and Law Association (ILLA) (2023).
- **Authorship attribution across right-wing manifestos.**
Cristina Aggazzotti and Elizabeth Allyn Smith.
16th Biennial Conference of the International Association for Forensic and Legal Linguistics (IAFLL) (2023).
- **Disambiguating ‘ground truth’ for forensic applications in artificial intelligence.**
Cristina Aggazzotti and Elizabeth Allyn Smith.
Proceedings of the American Academy of Forensic Sciences Annual Conference, vol. 29, p. 380 (2023).
- **A comparison of two computational approaches to analyzing last words.**
Cristina Aggazzotti.
Proceedings of the American Academy of Forensic Sciences Annual Conference, vol. 28, p. 483 (2022).
- **A quantitative comparison of last words to related text genres.**
Cristina Aggazzotti.
Proceedings of the American Academy of Forensic Sciences Annual Conference, vol. 28, p. 486 (2022).
- **A quantitative assessment of last words using Suicide Note Assessment REsearch (SNARE).**
Cristina Aggazzotti.
Proceedings of the American Academy of Forensic Sciences Annual Conference, vol. 27, p. 370 (2021).
- **Reflexives and reciprocals in synchronous tree adjoining grammar.**
Cristina Aggazzotti and Stuart M. Shieber.
Proceedings of the 13th Int’l Workshop on Tree Adjoining Grammars and Related Formalisms, pp. 31–42 (2017).
- **Communication strategies to minimize bias and strengthen scientific foundations in forensic science.**
Cristina Aggazzotti, Katherine Ramsland, Carole E. Chaski, Elizabeth Allyn Smith, and Roderick T. Kennedy.
Proceedings of the American Academy of Forensic Sciences Annual Conference, vol. 23, p. 38 (2017).
- **Confirmation bias and metalinguistic awareness.**
Carole E. Chaski, Elizabeth Allyn Smith, and Cristina Aggazzotti.
Proceedings of the American Academy of Forensic Sciences Annual Conference, vol. 22, p. 902 (2016).

Teaching and Mentoring Experience

Prison Tutor and Justice-involved Youth Mentor

THE PETEY GREENE PROGRAM (SCI CHESTER; PHILLYCONNECT)

Philadelphia, PA

Jan. 2025 – Present

- Tutor incarcerated students for the GED or their associate’s degree; academically mentor system-impacted youth

Research Mentor

POLYGENE

Remote

Sep. 2022 – Sep. 2023

- Mentored high school students through their first research project, resulting in papers and symposium presentations
- Advised high school students on academics and possible career paths based on their interests and goals

Math/SAT Tutor and College Coach

CARNEGIE PREP, CAMBRIDGE COACHING, INVICTUS

Remote

Nov. 2021 – Aug. 2023

- Devised lesson plans and SAT test prep guides; tutored for the SAT and for math from elementary to college level
- Guided students through application essay writing and college admissions from start to finish

Teaching Fellow

HARVARD UNIVERSITY

Cambridge, MA

Aug. 2016 – May 2018

- Assisted in teaching two intro linguistics classes (Distinction in Teaching Award for each) and an intro syntax class
- Created curricula for and taught as the primary instructor tutorials on computational linguistics and forensic linguistics

Prison Tutor and Careers Class Assistant

SUFFOLK COUNTY HOUSE OF CORRECTIONS; BOSTON COMM. CORRECTIONS CENTER

Boston, MA

Feb. 2015 – May 2016

- Taught English to Spanish-, Portuguese-, and Mandarin-speaking inmates; tutored men and women on probation/parole for the GED and assisted in a careers class creating resumes and helping with job applications

Tutor and College Consultant

SELF-EMPLOYED

Newbury Park, CA

Sep. 2013 – Aug. 2014

- Tutored 30 students for the SAT, ACT, and all high school subjects; >300pt avg increase on SAT, 3pt avg increase on ACT
- Counseled 5 students through college selection, applications, resumes, and essays with successful admissions for all

Grants and Awards

July 2025	Demonstration Project Award , Johns Hopkins University's Data Science and AI Institute
Aug. 2018	Research grant , Harvard Linguistics Department
April 2018	Research grant , Harvard Institute for Quantitative Social Science
Sep. 2017	Travel grant , Harvard Institute for Quantitative Social Science
2016, 2017	Conference grant , Harvard Linguistics Department
2016, 2017	Distinction in Teaching Award , Bok Center for Teaching and Learning

Professional Memberships

Society of Woman Geographers	2025 – Present
NIST's OSAC for Forensic Science , Speaker Recognition	2024 – Present
Association for Computational Linguistics	2024 – Present
International Association for Forensic and Legal Linguistics	2022 – Present
International Language and Law Association	2022 – Present
American Academy of Forensic Sciences , Trainee Affiliate Member, Engineering & Applied Sciences	2022 – 2023

Reviewing

Books	Research Methods in Applied Linguistics
Journals	Scientific Data (Nature), Cognitive Computation, Frontiers in Communication
Conferences	ICASSP

Skills

Computer	Python, GitHub, $\LaTeX$, MS Excel; R, MATLAB, SPSS, OCaml (prior experience)
Linguistics	Computational linguistics, syntax, semantics, corpus annotation and statistics, tree adjoining grammar
Language	English (native), Italian (proficient), Spanish (intermediate), French (reading knowledge)
Soft skills	Adaptable, good communicator, detail-oriented, problem solver, team player, quick learner, enthusiastic

Mentoring and Leadership

Board Member , Society of Woman Geographers	2026 – Present
Faculty Advisor , JHU North American Computational Linguistics Open Competition Committee	2025 – Present
Mentor for high school research projects , Polygence	2022 – 2023
Mentor for incoming graduate students , Harvard Linguistics Department	2015 – 2018
Assistant Head Tutor , Harvard Linguistics Department	2017 – 2018
Language Universals Colloquium Co-organizer , Harvard Linguistics Department	2015 – 2017
Captain , Harvard Boxing Team	2016 – 2017

Invited Talks

- “Authorship Attribution”
Guest lecture in Linguistics 412: Language and the Law, University of Southern California
Los Angeles, CA
Nov. 2024, Apr. 2022
- Research and Graduate School Panel
Panelist for BLISS/SHARP programs, Harvard University
Cambridge, MA
July 2020, July 2018
- Linguists in Industry Panel
Panelist, Harvard University Linguistics Department
Cambridge, MA
Feb. 2019
- “Forensic Linguistics”
Guest lectures in Linguistics 83 and Linguistics 101, Harvard University
Cambridge, MA
Dec. 2018, Apr. 2017, Apr. 2016
- “When words deceive you—Finding clues to justice through linguistics”
Speaker in the Continuing Legal Education program, New Mexico State Bar
Albuquerque, NM
Oct. 2017

Interests

- Traveling (76 countries, 47 U.S. states), sailing, running, learning new skills/hobbies, sports, music, knitting/crocheting